Topology-resolved quantum volume

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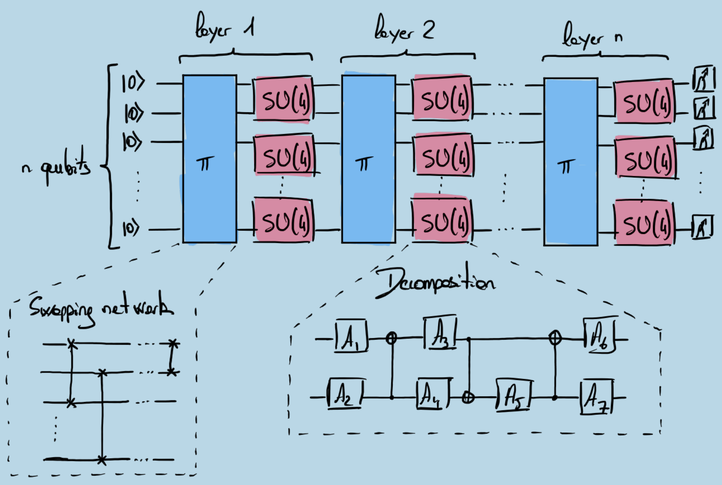
Problem

- NISQ devices have useful scale, but are limited by noise and lack fault tolerance.
- Performance depends on qubit quality, connectivity, and compilation overhead.
- Therefore, system-level benchmarks are needed in addition to gate-level metrics.

One such benchmark is quantum volume, which assesses only a limited subset of best-quality qubits within a quantum processor. We study how quantum volume depends on *where* a circuit is placed on a chip, not only on the chip as a whole.

Quantum volume

Quantum volume (QV) is a **system level benchmark**. The QV is a single number figure of merit used to evaluate universal gate-based quantum computers.



The QV circuit [1]

If the quantum computer samples a distribution from the **heavy set of outputs** (simulated classically) with probability $> 2/3$, it validates the corresponding quantum volume score of 2^n. The confidence interval for this evaluation is set to 2σ (97.7%).

Heavy set of outputs

Let $\mathbf{Q}$ be a random circuit drawn from a suitable ensemble acting on n qubits. The quantum state after executing the circuit is denoted $|\psi\rangle$. Each possible output state $x \in \{0,1\}^n$ is measured with probability $|\langle x|\psi\rangle|^2$. The set of output states with a probability greater than the median constitutes the heavy set of outputs associated with the quantum circuit $\mathbf{Q}$.

References

[1] Quantum Benchmark Zoo: *Quantum Volume*. Available at: `https://quantumbenchmarkzoo.org` (accessed 2026-04-12).

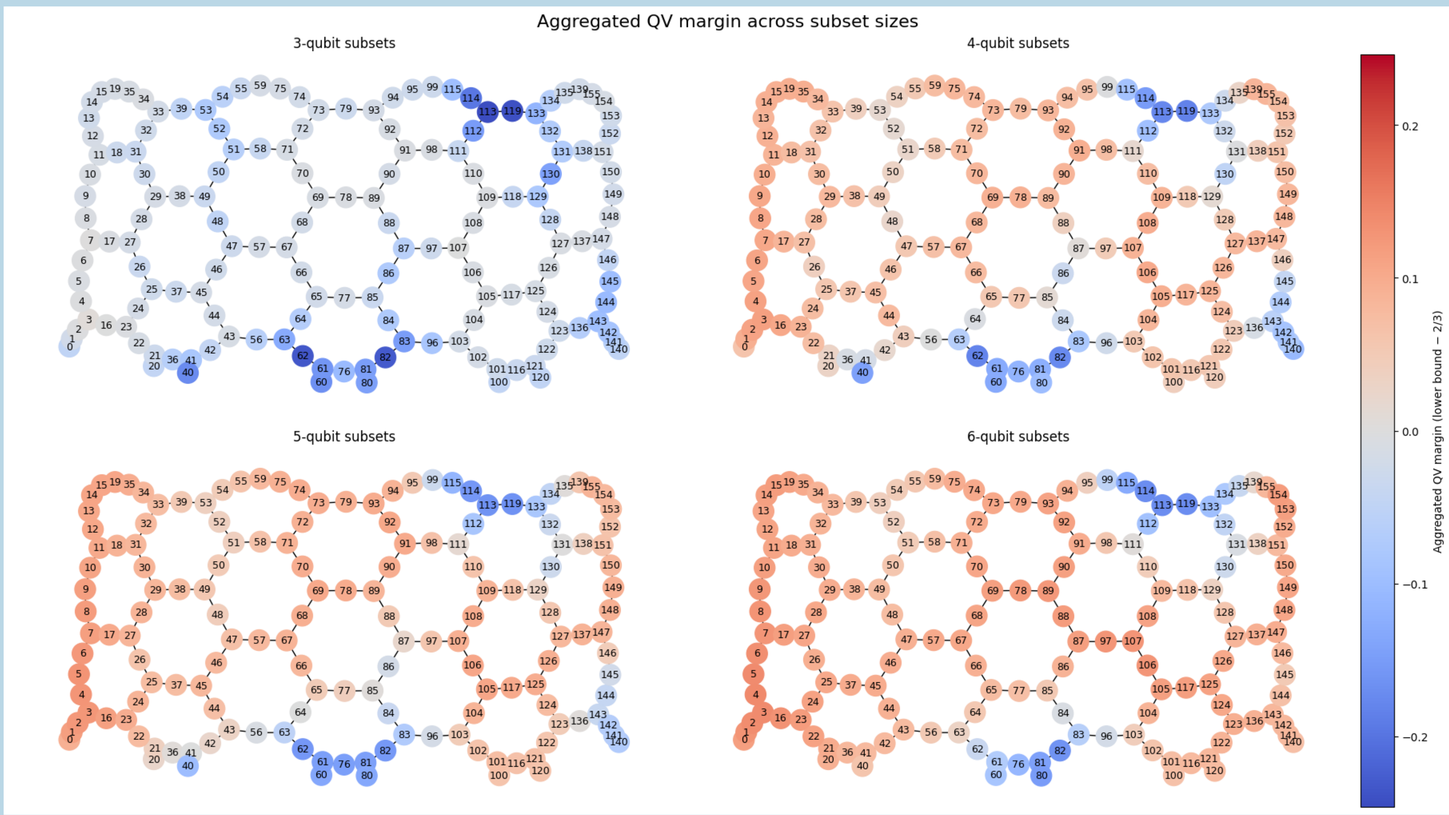
Acknowledgements

TThis work was supported by the University of Latvia student research funding, aimed at promoting the involvement of 1st and 2nd year bachelor-level and short-cycle students in research activities at the Faculty of Science and Technology (EZTF).

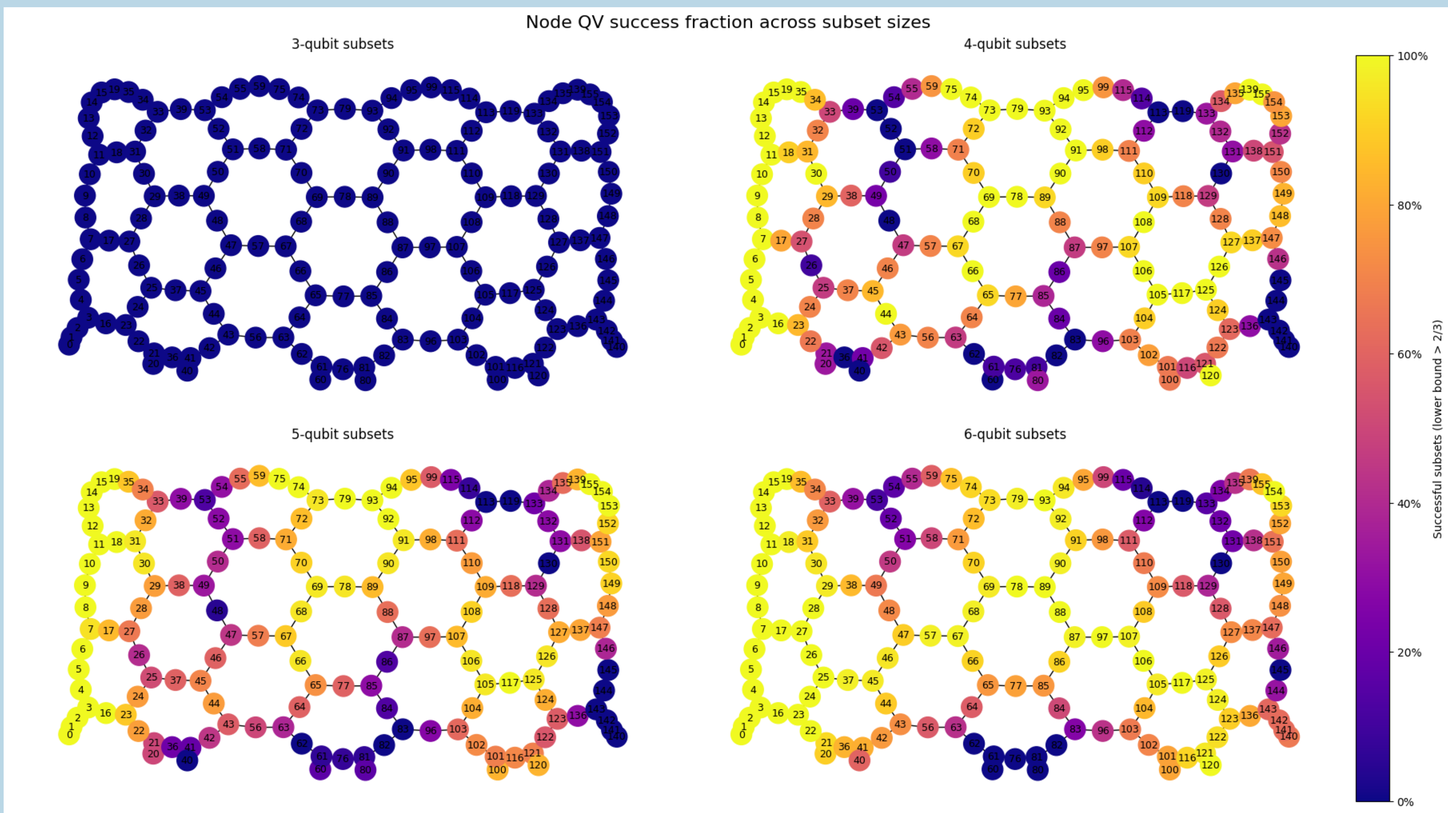
Research methodology

We benchmarked QV on the IBM Marrakesh device using calibration data and the coupling graph from Qiskit. First, we construct a noisy quantum simulator from the extracted data. Then we enumerate all connected subgraphs of size $n \leq 6$ and run QV circuits on each subgraph. Finally, we compute the heavy output probability and confidence for each placement.

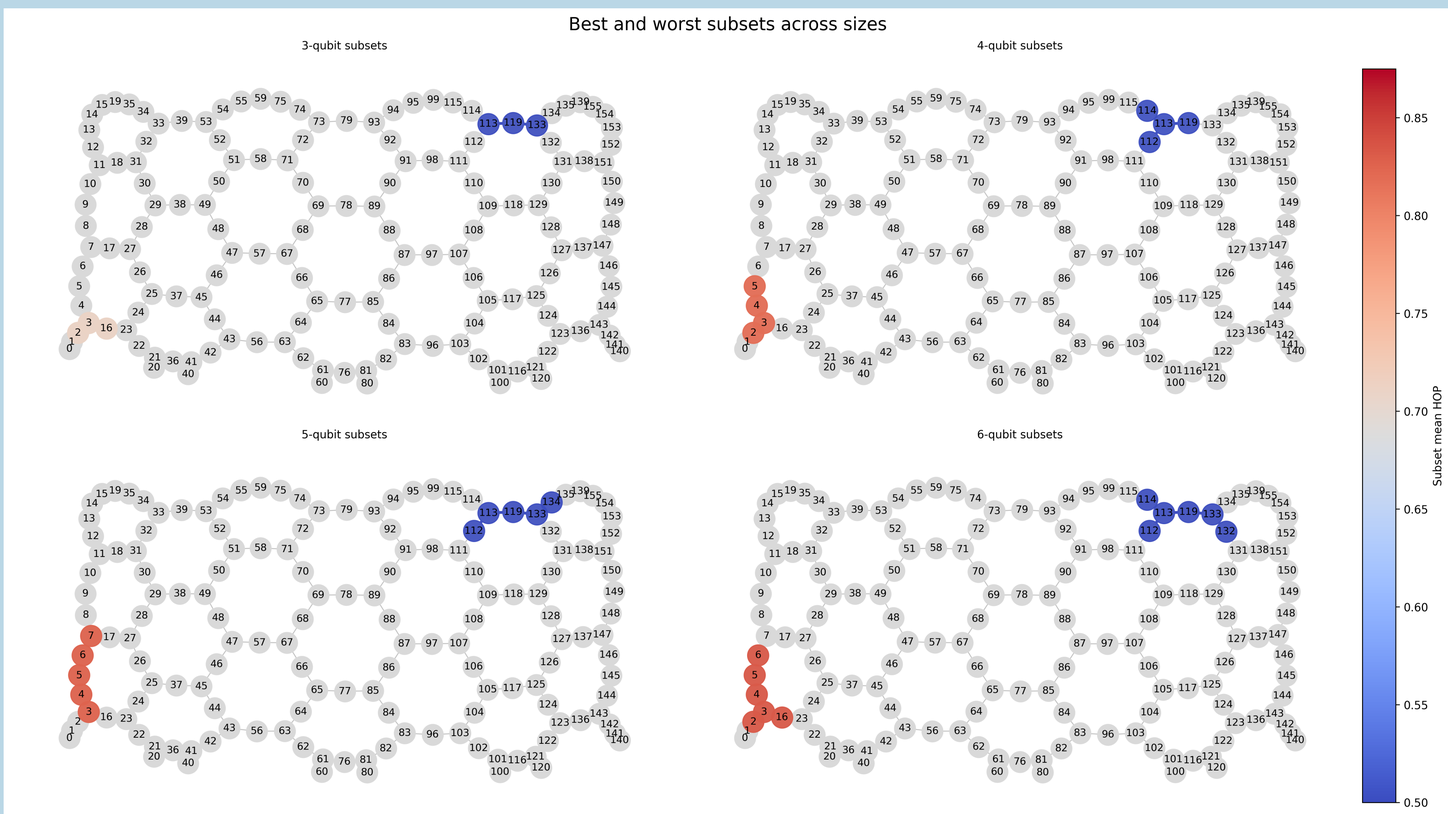
Productivity of qubit subsets varies significantly



Aggregated QV margin across subset sizes



Success fraction of QV circuits across subset sizes



Best and worst performing subsets across subset sizes